
FIT
FitFlex

FITFLEX -DATABASE DESIGN

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INTRODUCTION OF FITFLEX

Fit flex is Canada's largest retailer of sporting goods and equipment as same as Sport check. The company offers a diverse range of products, catering to sports and active lifestyles for men, women, and children. These include athletic footwear, fitness equipment, team sports gear, and outdoor adventure supplies.

OVERVIEW: Fitflex is an e-commerce store specializing in selling fitness-related products such as gym equipment, activewear, supplements, and accessories. The platform serves as a one-stop shop for customers, offering a seamless shopping experience with features like personalized recommendations, order tracking, and customer support.

Mission Statement

To design a robust, efficient, and secure database system for Fit Flex that allows data-driven decision-making and improves overall business performance.

A good database is helping to maintain data of company and help to know about customer preference, market trends and perform strategic like price and product in sports market.

Three main objectives for fitflex :

1. Design and implement a normalized database schema.

2. Generate reports and dashboards to support business decisions.

3. Ensure reliable, structured, and secure storage of customer, product, and transaction data.

Business Rules:

1. **Customer Rules**:

    - Every customer should have unique identifier which create uniqueness in details of customer.
    - Customer information should have information like firstname, lastname, email, phone and address.
    - Customer contact details must have firstname and lastname or valid email id.

2. **Employee Rules**

    - Every employee should have unique identifier which create uniqueness in details of employee.
    - Employee information should have information like name, position, email, storeId and address.
    - Employee must have email and valid storied detail.

3. **Product**

- Every product should have unique identifier which create uniqueness in details of product.
- Product information should have information like name, category, price, stock quantity.
- Product must have category and name of the product.

4. **Order**

- Every order should have unique identifier which create uniqueness in detail of order.
- Order information should have information like customerID, productID, order date, storeId.
- Order must have customerID, productID and storeID.

5. Store

- Every store should have unique identifier which create uniqueness in detail of store.
- Order information should have information like name of the store and location.
- Store must have name and location.

Subjects:

1. **Stockouts**:
   FitFlex sometimes runs out of important products, which can upset customers and slow down operations. Fixing this issue is important to keep things running smoothly and meet customer needs.

2. **Overstocking**:
   FitFlex also deals with having too many of certain products, which increases costs and can lead to losses. Better inventory management can help reduce waste and save money.

3. **Centralized System:**
   FitFlex wants to create one central system to handle operations, inventory, customer data, and sales in one place. This will save time, reduce errors, and help the company work more efficiently.

4. **Record Accuracy**:
   Right now, FitFlex uses manual data entry, which often leads to mistakes. Using an automated system would make the data more accurate and reliable.

5. **Decision-Making**:
   FitFlex makes decisions based on combined data from central systems, real-time feedback, and past trends. This helps the company adapt to market changes, choose the right products, and plan better strategies.

Tables:

Customer Table:

Store information about customer firstname, lastname,email, phone, address.

Employee Table:

Record information about employee firstname, lastname, email,phone, role.

Product Table:

Manage information about product name, description, price, categoryid.

Product category Table:

Store information about product category.

SalesOrder Table:

Track order information about customerId, employeeid, storeid, orderdate, productId, saleprice, quantity, orderstatus, paymentid.

Store Table:

Store information about storename, address and phone.

Paymentmode Table:

The method used to make the payment for an order. It can be Online or Offline.

OrderStatus Table:

The current stage of an order in the process. It can be Pending , Shipped, Delivered or Canceled .

List of attributes:

Customer Table:

1. **Customer_id:** is the unique identifier that describe each customer in the fit flex database. Customer_id is operates as the primary key into the database table that help to data retrieve and information handling.
2. **Customer_firstname**: The first name of the customer. This helps personalize interactions and serves as a reference for the user.
3. **Customer_lastname**: The first name of the customer. This helps personalize interactions and serves as a reference for the user.
4. **Customer_email:** The email associated with the customer.It helps in unique classification and for precise email identification
5. **Customer_phonenumber:** The contact phone number off the customer. This may include the country code and is used for communication purposes.
6. **Customer_address:** The complete address of the customer, including street, city, state, etc. It is used for delivery, billing, or verification purposes

Employee Table:

1. **Employee_id:**A unique identifier for each employee. This is typically the primary key and ensures that every employee has a unique reference in the database.
2. **Employee_firstname:** The first name of the employee, used for identification and internal communication purposes.

3. **Employee_lastname:** The last name of the employee, used for identification and internal communication purposes.
4. **Employee_email:** The email address of the employee, used for professional communication and system login credentials.
5. **Employee_phonenumber:** The employee phone number off the employee. This may include the country code and is used for communication purposes.
6. **Employee_role**: The job title or role of the employee within the fitflex. Helps to classify roles and responsibilities.

Product Table:

1. **Product_id:** A unique number or code assigned to each product to identify it in the system.
2. **Product_name:** The name of the product, describing what it is.
3. **Product_price:** The cost or selling price of the product.
4. **Product_categoryid:** The category or type of the product, used to group similar items together.

ProductCategory Table:

1. **Category_id:** A unique number or code assigned to each product category to identify it in the system.
2. **Category_name:** The name of the category, describing the type or group of products it includes.

Salesorder Table:

1. **Order_id:** A unique number or code assigned to each order to identify it in the system.
2. **Customer_id:** The unique number or code of the customer who placed the order, linking the order to a specific person
3. **Employee_id:** A unique number or code assigned to an employee, identifying the staff member handling the order.
4. **Store_id:** The unique number or code of the FitFlex store where the order was placed or processed.
5. **Order_date:** The date when the order was placed. Helps to track order history and timelines.
6. **Product_id:** The unique number or code of the product included in the order, identifying what was purchased.
7. **Product_Quantity:** The number of units of the product included in the order.
8. **Order_statusId:** The current state of the order (e.g., Pending, Shipped, Delivered, Canceled).
9. **Payment_id:** A unique number or code assigned to the payment transaction for tracking and verification.

Store Table:

1. **Store_id:** unique identity of the store.
2. **Store_name:** The name of the FitFlex store, used to identify it.
3. **Store_address:** The address or area where the store is located.

4. **Store_phone:** The contact phone number of the FitFlex store for customer inquiries and communication.

Payment mode Table

1. **Payment_id:** A unique number or code assigned to each payment transaction for identification.
2. **Payment_type:** The method of payment used (e.g., Cash, Online).

Order Status table

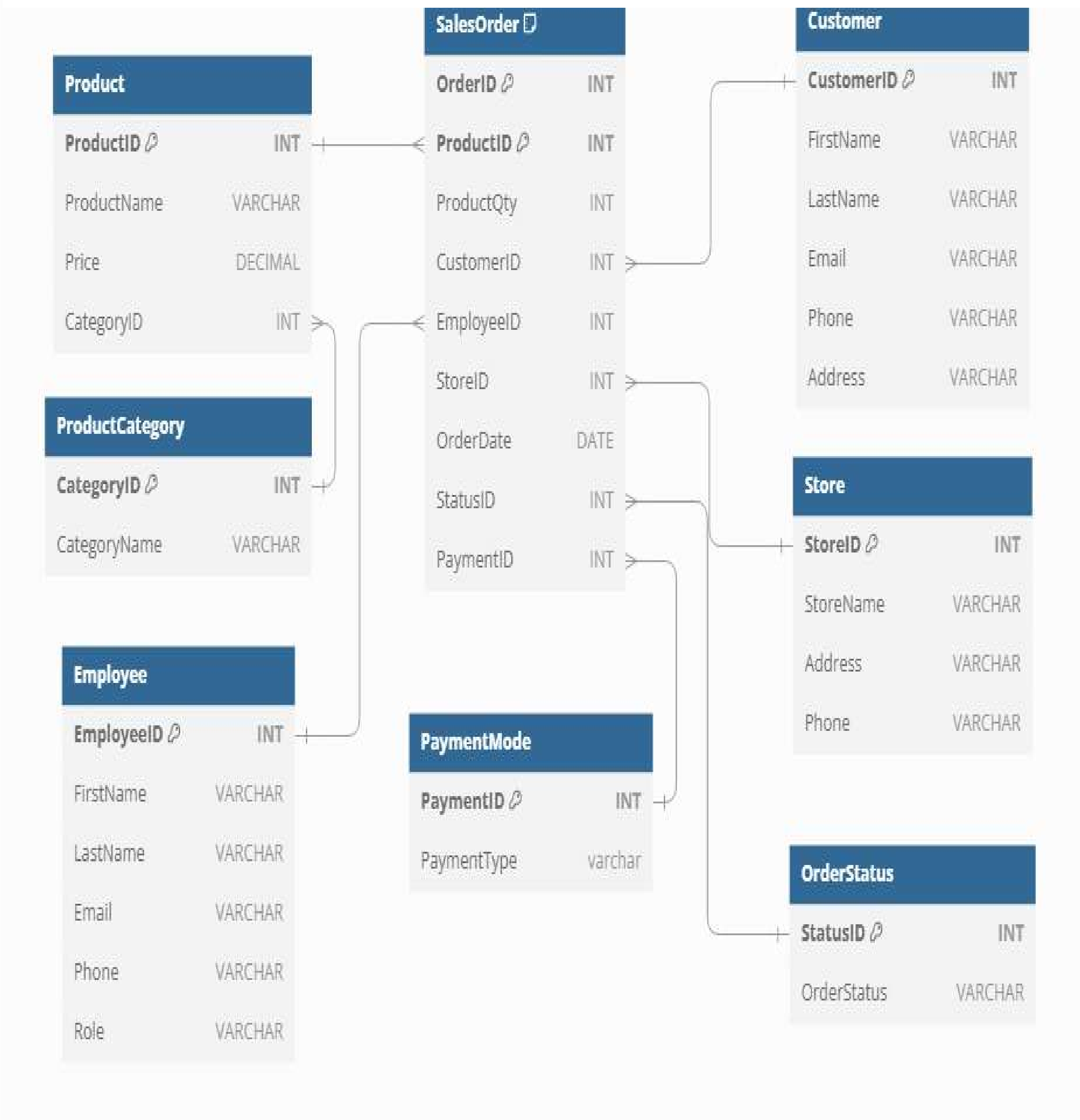
1. **Status_id:** A unique number or code assigned to each order status for identification.
2. **Order_status:** The current state of the order (e.g., Pending, Shipped, Delivered, Canceled).

Relationship for Fitflex

One-to-Many Relationships:

✓ **Customers → SalesOrders** (One customer can place multiple orders)
✓ **SalesOrders → Products** (One order can have multiple products)
✓ **SalesOrders → Store** (One order is placed at a single store, but a store can process multiple orders)
✓ **ProductCategory → Products** (One category can have multiple products)
✓ **PaymentMode → SalesOrders** (One payment method can be used in multiple orders)
✓ **Store → SalesOrders** (One store can have multiple orders)
✓ **OrderStatus → SalesOrders** (One order status can be assigned to multiple orders)

Entity Relationship (ER) Diagram



Data Dictionary:

I created a database for fitflex company, which can help them to improve the data accuracy and maintain fitflex data.

I created eight tables in database like customer, employee, product,productcategory, salesorder, store, paymentmode and orderstatus table and set key which help to retrieve data from different table .it is best for data analysis.

Customer Table:

Fields	Key	Datatype
Customer_id	Primary key	integer
Customer_firstname		varchar
Customer_lastname		varchar
Customer_email		varchar
Customer_phone		varchar
Customer_address		varchar

Employee Table:

Fields	Key	Datatype
Employee_id	Primary key	integer
Employee_firstname		varchar
Employee_lastname		varchar
Employee_email		varchar
Employee_phone		integer
Employee_role		varchar

Product Table

Fields	Key	Datatype
Product_id	Primary key	integer
Product_name		varchar
Product_prize		decimal
Product_category_id		integer

ProductCategory Table

Fields	Key	Datatype
Category_id	Primary key	integer
Category_name		varchar

Salesorder Table

Fields	Key	Datatype
Order_id	Primary key	integer
Customer_id	Foreign Key	integer
Employee_id	Foreign Key	integer
Store_id	Foreign Key	integer
Order_date		date
product_id	Foreign Key	integer
Quantity		integer
status_id		integer
Payment_id		integer

Store Table

Fields	Key	Datatype
Store_id	Primary key	integer
Store_name		varchar
Store_address		varchar
Store_phone		varchar

Paymentmode Table

Fields	Key	Datatype
Payment_id	Primary key	integer
Payment_type		varchar

Order status Table

Fields	Key	Datatype
Status_id	Primary key	integer
Orderstatus		varchar

Views and Query

- **Order Details –** Customer Receipt
- **Hot Selling Product –** Group by product. Sum of sales by product.
- **Loyal Customer –** Group by customer. Sum of sales by customer.
- **Best Performing Employee –** Group by employee.

Views for customer Receipt

```sql
CREATE VIEW OrderDetails AS
SELECT
    so.OrderID,
    CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName,
    CONCAT(e.FirstName, ' ', e.LastName) AS EmployeeName,
    s.StoreName,
    os.OrderStatus,
    pm.PaymentType,
    p.ProductName,
    so.ProductQty,
    (p.Price * so.ProductQty) AS ProductTotal,
    (SELECT SUM(p2.Price * so2.ProductQty)
     FROM SalesOrder so2
     JOIN Product p2 ON so2.ProductID = p2.ProductID
     WHERE so2.OrderID = so.OrderID) AS OrderTotal
FROM
    SalesOrder so
JOIN
    Customer c ON so.CustomerID = c.CustomerID
JOIN
    Employee e ON so.EmployeeID = e.EmployeeID
JOIN
    Store s ON so.StoreID = s.StoreID
JOIN
    OrderStatus os ON so.StatusID = os.StatusID
JOIN
    PaymentMode pm ON so.PaymentID = pm.PaymentID
JOIN
    Product p ON so.ProductID = p.ProductID;

SELECT * FROM OrderDetails WHERE OrderID = 2;
```

Results Messages

	OrderID	CustomerName	EmployeeName	StoreName	OrderStatus	PaymentType	ProductName	ProductQty	ProductTotal	OrderTotal
1	2	Dwight Schrute	Jane Smith	Uptown Fitness	Completed	Credit Card	Resistance Bands	3	59.97	2609.76
2	2	Dwight Schrute	Jane Smith	Uptown Fitness	Completed	Credit Card	Running Shoes	7	629.93	2609.76
3	2	Dwight Schrute	Jane Smith	Uptown Fitness	Completed	Credit Card	Hiking Backpack	9	1169.91	2609.76
4	2	Dwight Schrute	Jane Smith	Uptown Fitness	Completed	Credit Card	Fitness Tracker	5	749.95	2609.76

Purpose: want to show the customer the details of their order

Tables used: SalesOrder, Customer, Employee, Product, Store, OrderStatus, PaymentMode

View Name: OrderDetails

Fields Used: CustomerName, EmployeeName, StoreName, OrderID, OrderStatus, PaymentType, ProductName, ProductQty, OrderTotal

Calculated Fields: ProductTotal, OrderTotal, Customer Name, EmployeeName

Views for top 5 hot selling products

```sql
CREATE VIEW HotProducts AS
SELECT
    p.ProductID,
    p.ProductName,
    SUM(so.ProductQty) AS TotalQuantitySold,
    SUM(so.ProductQty * p.Price) AS TotalSales
FROM
    SalesOrder so
JOIN
    Product p ON so.ProductID = p.ProductID
GROUP BY
    p.ProductID, p.ProductName
ORDER BY
    TotalSales DESC
LIMIT 5;

SELECT * FROM HotProducts
```

Results   Messages

	ProductID	ProductName	TotalQuantitySold	TotalSales
1	1	Treadmill	28	27999.72
2	10	Elliptical Machine	15	8999.85
3	7	Hiking Backpack	28	3639.72
4	8	Fitness Tracker	21	3149.79
5	2	Dumbbell Set	15	2999.85

Purpose: Identify the top 5 selling products

Tables used: SalesOrder, Product

View Name: Hot Selling Products

Fields Used: ProductID, ProductName

Calculated Fields: TotalQuantitySold, TotalSales

Best performing employees

```sql
SELECT
    e.EmployeeID,
    CONCAT(e.FirstName, ' ', e.LastName) AS EmployeeName,
    SUM(so.ProductQty * p.Price) AS TotalSales
FROM
    SalesOrder so
JOIN
    Employee e ON so.EmployeeID = e.EmployeeID
JOIN
    Product p ON so.ProductID = p.ProductID
GROUP BY
    e.EmployeeID, EmployeeName
ORDER BY
    TotalSales DESC
LIMIT 5;
```

Results   Messages

	EmployeeID	EmployeeName	TotalSales
1	6	David Brown	13289.71
2	3	Mike Taylor	10319.78
3	8	Linda Garcia	8599.75
4	1	John Doe	5819.83
5	5	Sarah Lee	4489.71

Purpose: Identify the top 5 performing employees

Tables used: SalesOrder, Product, Employee

Fields Used: EmployeeID, eFirstName, eLastName

Calculated Fields: EmployeeName, TotalSales

Conclusion

The FitFlex database is designed to manage important details about customers, employees, products, stores, and orders. With clear tables and useful attributes like IDs, names, and locations, it makes tracking and operations easier. This helps FitFlex deliver better service, manage inventory, and run stores more smoothly. The database is a solid base for future growth and success.